

Password-Protected Parking System Controller

VHDL Design and Implementation on Nexys A7-100T FPGA

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Abstract

This document describes the design and implementation of a password-protected parking system controller, developed in VHDL and verified on a Digilent Nexys A7-100T FPGA board. The system manages vehicle entry and exit, tracks available parking capacity, and restricts entry to users who supply a correct access code via on-board switches. Status is reported through a seven-segment display and LED indicators.

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1 Project Overview

This project implements a password-protected parking system controller in VHDL, designed and tested on an FPGA board. The system controls vehicle entry and exit, maintains a running count of available parking spaces, and prevents new entries once the lot is full.

The parking lot has a maximum capacity of **15 spaces**. The controller maintains an internal count of occupied spaces and computes the number of available spaces as:

$$\text{available spaces} = 15 - \text{occupied spaces} \quad (1)$$

The available space count is shown on a seven-segment display. The two rightmost digits display the number of free spaces, while the leftmost digit indicates password verification status:

A – password accepted

E – wrong password entered

2 Access Control

Vehicle entry is protected by a simple password mechanism. The password is entered using four on-board switches, which simulate an RFID/card-based access input.

2.1 Authorized Code

In the VHDL source, the authorized code is defined as a 4-bit constant compared directly against the switch vector `SW(3 downto 0)`:

```
1 constant AUTHORIZED_CODE : STD_LOGIC_VECTOR(3 downto 0) := "1010";
```

This means the required logic levels are `SW3 = 1`, `SW2 = 0`, `SW1 = 1`, `SW0 = 0`. On the physical Nexys A7 board, switch `SW0` is the rightmost switch on the panel; physically setting switches `SW0`–`SW3` to match this pattern (`SW0` down, `SW1` up, `SW2` down, `SW3` up) satisfies the comparison in hardware.

2.2 Verification Procedure

Before a vehicle may enter, the user must:

1. Set the four switches `SW[3:0]` to the access code.
2. Press `BTNL` (Verify Password).

If the code is correct *and* the lot is not full, the display shows **A** and one vehicle is permitted to enter on the next press of `BTNU` (Entry). If the code is incorrect, the display shows **E** and entry is blocked until verification succeeds.

3 Controls and Indicators

3.1 Push Buttons

3.2 LED Status Indicators

3.3 Debounce Logic

All four push buttons are routed through a dedicated `debounce_pulse` module before being used by the control logic, in order to suppress false multiple-trigger events caused by mechanical

Button	Function
BTNC	Reset
BTNL	Verify password
BTNU	Entry
BTND	Exit

Table 1: Push button assignments

LED	Indicates
LED0	Space available
LED1	Parking full
LED2	Password accepted
LED3	Password rejected

Table 2: LED status assignments (LED[15:4] unused, held low)

switch bounce. Each instance double-synchronizes the input and requires the signal to remain stable for a configurable number of clock cycles (`STABLE_COUNT`, default 2,000,000 cycles at 100 MHz $\approx$ 20 ms) before registering a clean single-cycle pulse.

4 System Architecture

The design is partitioned into two VHDL entities:

- `debounce_pulse` – a reusable synchronizer/debounce module instantiated four times (entry, exit, reset, verify buttons).
- `parking_top` – the top-level entity containing the password comparison logic, occupancy counter, seven-segment multiplexing, and LED status outputs.

4.1 Occupancy and Display Logic

The occupancy counter increments on a valid, password-authorized entry pulse and decrements on an exit pulse, saturating at 0 and 15. The available count is split into tens and ones digits and multiplexed across the seven-segment display anodes using a refresh counter, cycling through three display positions: ones digit, tens digit, and the access-status character (A/E).

4.2 Source Code

debounce_pulse

```

1 entity debounce_pulse is
2     generic (
3         STABLE_COUNT : natural := 2000000
4     );
5     Port (
6         clk      : in  STD_LOGIC;
7         btn_in   : in  STD_LOGIC;
8         pulse    : out STD_LOGIC
9     );
10 end debounce_pulse;

```

parking_top (port interface)

```

1 entity parking_top is
2     generic (
3         DB_COUNT : natural := 2000000
4     );
5     Port (
6         CLK100MHZ : in STD_LOGIC;
7         BTNU : in STD_LOGIC; -- Entry
8         BTND : in STD_LOGIC; -- Exit
9         BTNC : in STD_LOGIC; -- Reset
10        BTNL : in STD_LOGIC; -- Verify password
11        SW : in STD_LOGIC_VECTOR(3 downto 0); -- Password input
12        LED : out STD_LOGIC_VECTOR(15 downto 0);
13        CA, CB, CC, CD, CE, CF, CG, DP : out STD_LOGIC;
14        AN : out STD_LOGIC_VECTOR(7 downto 0)
15    );
16 end parking_top;

```

Full source listings for both entities, including the password comparison process, occupancy counter, and seven-segment encoding functions, are included in the project repository under `src/`.

5 Pin Assignments (Nexys A7-100T XDC)**5.1 Clock**

Signal	Pin	Standard
CLK100MHZ	E3	LVC MOS33 (100 MHz)

5.2 Password Switches

Signal	Pin	Standard
SW[0]	J15	LVC MOS33
SW[1]	L16	LVC MOS33
SW[2]	M13	LVC MOS33
SW[3]	R15	LVC MOS33

5.3 Push Buttons

Signal	Pin	Function
BTNU	M18	Entry
BTND	P18	Exit
BTNC	N17	Reset
BTNL	P17	Verify password

Signal	Pin	Signal	Pin
LED[0]	H17	LED[8]	V16
LED[1]	K15	LED[9]	T15
LED[2]	J13	LED[10]	U14
LED[3]	N14	LED[11]	T16
LED[4]	R18	LED[12]	V15
LED[5]	V17	LED[13]	V14
LED[6]	U17	LED[14]	V12
LED[7]	U16	LED[15]	V11

Table 3: Only LED[0]–LED[3] are functionally used (see Section 3.2)

5.4 LEDs

5.5 Seven-Segment Display

Signal	Pin	Description
CA	T10	Segment A
CB	R10	Segment B
CC	K16	Segment C
CD	K13	Segment D
CE	P15	Segment E
CF	T11	Segment F
CG	L18	Segment G
DP	H15	Decimal point

Signal	AN[0]	AN[1]	AN[2]	AN[3]	AN[4]	AN[5]	AN[6]	AN[7]
Pin	J17	J18	T9	J14	P14	T14	K2	U13

Table 4: Seven-segment display anode pins (AN0 = ones digit, AN1 = tens digit, AN7 = status character)

6 Verification and Testing

The design was verified using the following flow:

1. **Functional simulation** – a VHDL testbench was used to exercise reset, password verification (correct and incorrect codes), entry/exit sequencing, and full-lot behavior.
2. **Synthesis and implementation** – the design was synthesized and implemented targeting the Nexys A7-100T using the constraints in Section 5.
3. **Hardware testing** – the bitstream was loaded onto physical FPGA hardware and verified against the button, switch, LED, and seven-segment behavior described above.

7 Hardware Concept (PCB)

A custom PCB concept supporting this design is in development in Altium Designer. The planned board includes:

- Password input switches and push buttons
- Status LEDs
- Seven-segment display
- FPGA interface connections
- Power supply circuitry
- Clock source
- Configuration/programming circuitry

This document represents a working draft and will be updated as simulation, hardware testing, and PCB design progress.